

Temporal confounding in evidence that psychedelics align brain activity with context

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Abstract

Though usually represented within one latent space, some psychedelic neural trajectories change their spatial and/or temporal geometry across environmental contexts. CEBRA (a contrastive embedding model) readily produces one low-dimensional geometry for these context-dependent, also interpreted as participant-shared, neural trajectories, but it consistently fails to establish their independence from acquisition order in these fixed-order recordings. Stoliker and colleagues trained CEBRA and a temporal attention-enhanced variational graph recurrent neural network (TAVRNN) to infer contextual organization of fMRI dynamics using a representation they consider a participant-shared correlate of acute experience and mindset change on the following day. However, their study lacks some critical safeguards required to evaluate order-independent contextual effects. We conclude that these tests show the reported advantage at the psilocybin visit to be an unreliable indicator of participant-shared alignment between brain activity and context. First, stricter participant-held-out evaluation of the public recordings reveals that mean balanced accuracy at baseline exceeds the near-chance mean at the psilocybin visit by 0.140, disputing the inference that context decoding generalizes by capturing a psilocybin-related strengthening of participant-shared alignment between brain activity and context. Moreover, we determine that fixed-order analysis treats four position-specific scan runs as environmental contexts, a fundamental ambiguity liable to distort conclusions. Finally, participant-level sensitivity analysis establishes that the reanalysis yields 576 CEBRA metric-behaviour association tests with negligible multiplicity-adjusted support and nine ostensibly network-attributed fixed-classifier losses with estimates different from corresponding refitted-classifier ones. On these grounds, we recommend adopting an observational account of the relation reported in healthy volunteers between participant-shared alignment of brain activity with context at the psilocybin visit and mindset change measured one day later, qualified by these evidential limitations, as a more limited and less mechanistically specific alternative to the proposed therapeutic interpretation of that relation.

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Participant-shared alignment of brain activity with context under psychedelics is a consequential hypothesis for interpreting patterns in neural trajectories that connect acute experience with changes in mindset one day later—that is, a proposed neural link between environmental context and psychological change¹. Stoliker and colleagues used CEBRA and a temporal attention-enhanced variational graph recurrent neural network (TAVRNN) in baseline and psilocybin-visit fMRI recordings to infer participant-shared alignment between brain activity and context that was stronger at the psilocybin visit and covaried with acute experience and mindset change one day later¹. We reassessed the public recordings from Stoliker and colleagues using out-of-sample validation procedures to control for temporal overlap and for serial dependence among frames acquired from one participant, and found no consistent evidence that the psilocybin visit strengthened brain–context alignment. Our reanalysis indicates that the released evidence does not substantiate Stoliker and colleagues’ claim, on the basis of these fixed-order scans, to have established participant-shared brain–context alignment or therapeutic relevance for that alignment.

The resulting risk of inflated decoding accuracy due to dependent observations in structured-data analyses has been acknowledged in methodological guidance³. When evaluating the influence of random partitioning on autocorrelated frames, most test frames are anticipated to neighbour a training observation (an adjacent frame) even in settings with no neural representation generalizable across scans. Stoliker and colleagues tested four runs for a decodable contrast in framewise brain dynamics due to context labels and their fixed acquisition positions within each visit-specific sequence of CEBRA-Time recordings (four conditions at positions 1–4; Methods section of the Article). Stoliker and colleagues applied a visit-wide (session-specific) CEBRA model (one fit across four concatenated runs) as well as support-vector classifiers based on fixed-order condition labels (rest, meditation, music and movie) and undocumented train–test assignments. They reported all 54 participants achieving decoding accuracy above the 0.25 chance level across four condition classes. Notably, the possible random-frame split that we evaluated would be predicted to yield extensive training–test adjacency (among held-out frames) within each session, creating uncertainty about whether context decoding is temporally generalizable.

We examine the central claim advanced in the Article, which was that CEBRA trajectories (latent representations, defined in three coordinates (3D)) displayed a latent geometry of condition separation and behavioural covariation across embedded states that was selectively strengthened by drug status and context (visit- and condition-level factors) (Fig. 1a; Methods). Stoliker and colleagues inferred this pattern by comparing neural recordings from the psilocybin visit with baseline recordings that did not include psilocybin exposure (baseline visit (no psilocybin)). Here we concentrate on the primary CEBRA classification of four context-labelled runs per visit (rest, meditation, music and movie) that occupied fixed acquisition positions (first to fourth), but the same time–context ambiguity also pertains to TAVRNN and network-replacement analyses that were conducted with the same acquisition sequence.

The established convention for evaluating predictions from depen-

dent data relies on blocked or group-held-out cross-validation (CV), which is understood as the separation of test observations before feature learning^{3,4}. In practice, a purge interval is often inserted to account explicitly for leakage from temporal autocorrelation³. Indeed, a related methodological report from the CEBRA developers used cross-session consistency to evaluate neural representations². Using participant-held-out validation guarantees that no frames from the held-out participant will be present in the training data. When we implemented participant-held-out validation, we observed no statistical support for above-chance accuracy at the psilocybin visit or for a positive paired advantage over baseline; the mean visit difference was -0.140 (Fig. 1b).

A further concern with the workflow is its treatment of each fixed-order run as a distinct context. This design does not allow for the one-to-one correspondence between context label and scan position imposed by the fixed run order. Differences among post-dose time points are an important source of variation in psychedelic neuroimaging^{5,6}. By treating every fixed-position run as a distinct context, the workflow ignores the complete confounding of condition with scan position and results in overstated evidence for context specificity. Accordingly, these findings might not apply to participants who are scanned under counterbalanced condition orders, failing to establish reliable contextual effects on shared brain representations. The most robust strategy for estimating contextual effects with repeated neuroimaging data addresses temporal confounding by using counterbalanced datasets that pool all condition-specific observations that were acquired at different scan positions. Alternatively, contextual effects in longitudinal neuroimaging datasets can be estimated using models that incorporate measured post-dose time to control for time-dependent drug effects⁵ or controlled-stimulus designs that are insensitive to low-level sensory variation while controlling for gaze-dependent visual responses⁷.

We reassessed the reported brain–behaviour associations using participants ($n = 57$) rather than frames (multiple observations per participant) as the analytical unit (Extended Data Fig. 1d–f). We found just 4 of 576 CEBRA–behaviour comparisons with nominal statistical evidence (without adjustment) (P below 0.05). The minimum multiplicity-adjusted q estimate among the 576 full- and proxy-cohort comparisons was 0.996.

The possibility that condition labels are recoverable from absolute acquisition time alone can be readily shown by comparing absolute-time decoding with a within-run-phase control. We separately decoded the condition labels (rest, meditation, music and movie) of the participant records from absolute time and within-run phase, and applied all four displayed validation schemes to both temporal feature sets. The range of mean accuracies characterizing absolute-time classification (absolute frame time alone) for the four displayed validation schemes (0.790–0.988 (participant averages)) was well above the corresponding within-run-phase range (0.232–0.250) (Fig. 1d). Notably, 100% of the 256 participant-level absolute-time estimates corresponded to a balanced accuracy greater than the 0.25 chance level.

However, we observe that nearest-neighbour time distances consistently undercut random-pair distances across validation regimes,

and subtraction of the circular-shift temporal null yields only modest residual visit differences. By contrast, multiplicity-adjusted com-

parisons between adjacent purge gaps pointed to no reliable change across the three pairs, as expected.

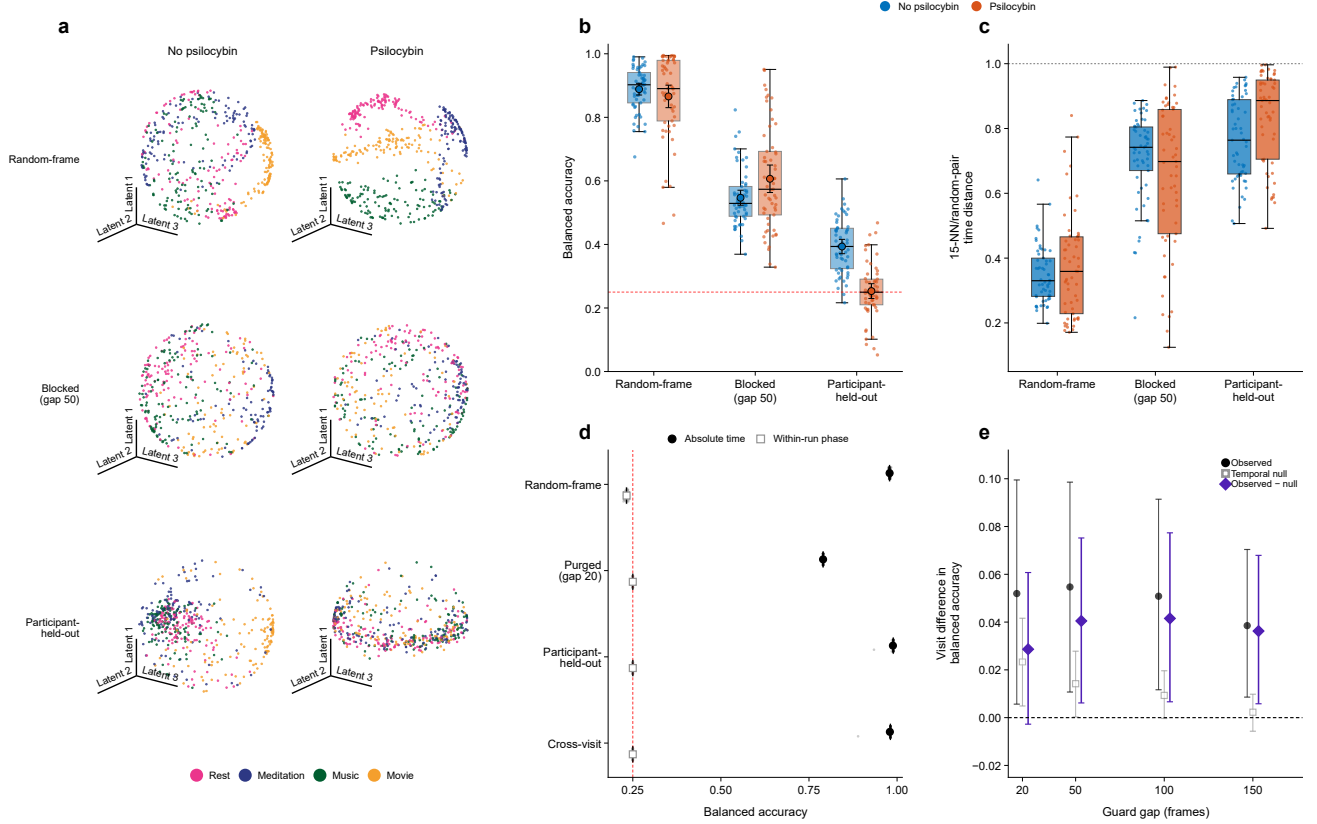


Fig. 1 | Validation-dependent organization of fixed-order CEBRA geometry under temporal controls. **a**, Condition-coloured CEBRA geometry (post-dose and baseline visits) selected by the archived cohort-medoid rule across the cohort for every validation regime (rows, regimes; columns, visits; up to 120 frames per condition; embeddings independently centred and rigidly rotated, then displayed under a common orthographic view and unit-sphere scale; axis orientation not comparable across fits). The full-cohort silhouette and temporal-neighbour distributions after these validation schemes are presented within Extended Data Fig. 1b,c. **b**, Visit-stratified balanced accuracy (no-psilocybin and psilocybin visits; dashed line, chance at 0.25) across 57 participants. Participant-held-out validation yielded no psilocybin accuracy advantage within participants during independent evaluation. **c**, 15-nearest-neighbour time-distance ratio (neighbour distance divided by

random-pair distance) estimated within every participant and thereafter summarized over the cohort, for every visit-by-validation stratum. **d**, Participant-averaged balanced accuracy for four displayed validation schemes (rows) and two temporal feature sets (absolute time, filled circles; within-run phase, open squares). Accuracy separation between absolute time and within-run phase at random-frame validation (top row) clearly persists after temporal or participant separation (remaining rows). **e**, Temporal-null subtraction leaves a modest visit-accuracy excess across guard intervals (0.029–0.042) for every guard comparison. Every small mark represents a participant (panels **b** and **c**, $n = 57$ per visit and regime; panel **d**, $n = 65$ for the initial three settings and $n = 61$ for cross-visit transfer), box glyphs and large symbols denote medians with quartile limits and means, respectively, and bars and whiskers are 95% confidence intervals and $1.5 \times \text{IQR}$, respectively.

Consequently, blocked validation does not furnish convincing evidence that visit identity alone supports reliable separation of context representations independent of acquisition order.

In their assessment of embedding stability, Stoliker and colleagues report that they repeated CEBRA fitting to assess variation among participant-specific embeddings, thus quantifying the influence of stochastic optimization. Yet, the Article presents condition-separability estimates for identical within-session observations across repeated model fits (Fig. 3e,f of Stoliker et al.). Moreover, by using time points that later enter evaluation (for example, neighbouring frames within a run), the embedding procedure that Stoliker and colleagues describe is an instance of data reuse that employs the same within-session observations (the four fixed-order runs) to construct the representation and to evaluate condition separability. Such reuse would give rise to apparently distinct condition clusters even if no order-generalizable context representation were present.

To assess whether network-replacement losses reflect fixed-classifier mismatch within the procedure under examination, we examined a fixed-versus-refitted classifier contrast ($n = 54$ participants; Methods) derived from hybrid embeddings of the Arti-

cle’s substitution analysis (eight single-network masks and one joint mask; Fig. 2a). Likewise, losses concentrated in visual and default-mode masks, as identified by the Article, lacked consistent selectivity among networks in either sensitivity analysis, in marked contrast to the fixed-classifier ranking. Refitting each network-substituted embedding with a hybrid-trained classifier produces a pronounced attenuation (5.3–31.0 percentage points) in apparent accuracy loss across all nine masks (Fig. 2a; descriptive fixed-coordinate distance ratios in panel **b**), showing that the analytical choice of retaining a fixed classifier after network substitution drives the estimates obtained from hybrid embeddings. Thus, the fixed-classifier substitution method used by Stoliker and colleagues for establishing network contributions (loss versus mask size: Spearman $\rho = 0.970$, two-sided unadjusted $P = 6.55 \times 10^{-5}$; panel **c** of Fig. 2; equal-size and signed joint contrasts in panels **d** and **e**) departs from mask-size-adjusted practice in this setting, and classifier refitting and size-matched comparisons result in substantially different conclusions about network specificity.

One might contend that confounding from the fixed condition order ought to have been detected by robustness analyses directed at accounting for changes in model specification across validation set-

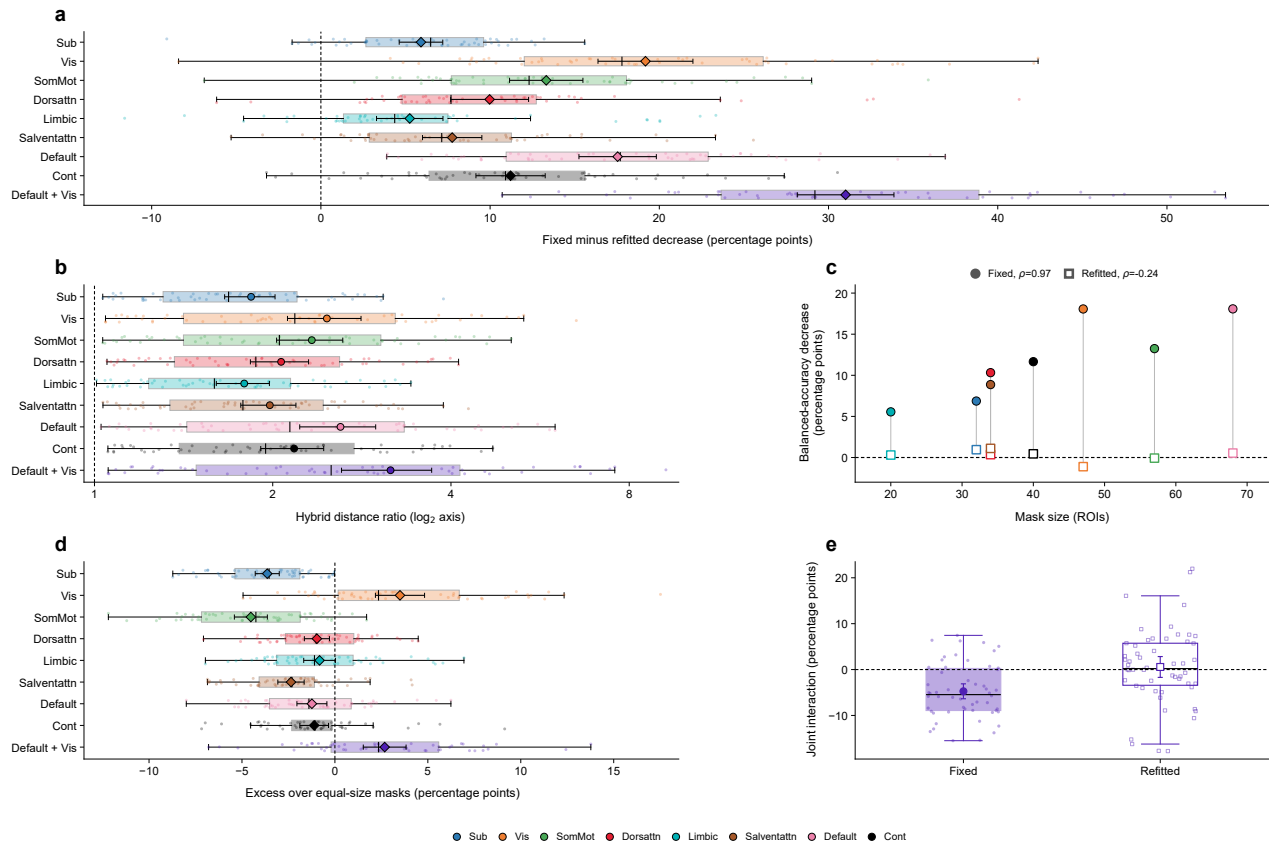


Fig. 2 | Network-replacement conclusions vary with estimator adaptation under distribution shift. **a**, Using network-substituted embeddings, for every participant, we transformed the difference between fixed-classifier and independently refitted-classifier loss at each replacement mask into an attenuation estimate, producing a profile spanning nine masks: eight single-network targets (network colours) and the joint Default + Vis target (purple). **b**, Fixed-coordinate hybrid-distance ratios (log₂ axis) for nine replacement masks, which descriptively indexed extrapolation in 54 participants (reference ratio, 1); their cohort means exceeded the within-psilocybin nearest-neighbour reference. **c**, For eight single-network masks, a Spearman analysis quantified the association between ROI count and balanced-accuracy decrease at the mask level. The mask-level Spearman association between ROI count and mean balanced-accuracy decrease ($n = 8$; fixed, $\rho = 0.970$, two-sided unadjusted $P = 6.55 \times 10^{-5}$; refitted, $\rho = -0.240$, two-sided

unadjusted $P = 0.568$) is displayed (panel c). **d**, Participant-level contrasts between fixed-seed-0 named-mask decreases (positive term) and mean decreases from 20 equal-size random masks (subtracted term; $n = 54$ per target) across all nine masks. Equal-size matching did not preserve either the original network ranking or uniformly positive excess (only Vis and Default + Vis had positive cohort means: 3.50 and 2.68 percentage points, respectively). **e**, A signed contrast mapped the joint Default + Vis decrease into joint minus the sum of separate Default and Vis decreases. Participant interactions were summarized as percentage-point means with 95% confidence intervals and displayed separately under the fixed and refitted classifiers (fixed, $-4.74 [-6.36, -3.11]$, subadditive; refitted, $0.56 [-1.71, 2.83]$, interval spanning zero). Every small mark represents a participant, distinguished by mask colour or estimator fill.

tings. We confirmed that our central conclusions do not hinge on the CEBRA validation scheme (using random-split, temporally blocked, participant-held-out and temporal-null estimates in Fig. 1b,d,e), the TAVRNN feature-density controls and 216 behavioural tests, the temporal purge gap (20, 50, 100 or 150 frames), choice of DCM mask, or the measure of latent geometry (ED Figs. 2d,e, 3a,b and 1b,c).

The participant-level fMRI and EEG analyses show that the most robust psilocybin-visit pattern comprises lower modularity⁸, attenuated eyes-closed-movie differences in GFC and alpha power, and higher LZ76 complexity^{9,10}, and indicate that these fixed-order observational contrasts do not isolate pharmacological action from visit order (ED Figs. 4 and 5). Yet, the Article's conclusions were not framed as limited visit associations or hypotheses about psychedelic brain-context relations. Instead, the Article claims to offer empirical support for participant-shared brain-context alignment, network-specific necessity and therapeutic relevance¹. Missing patient outcomes, randomized treatment comparisons and mediation analyses demonstrate that brain-context alignment is not established as a therapeutic mechanism by this healthy-volunteer study. Such interpretations could, for instance, prompt researchers to mistake acquisition-order effects for psychedelic mechanisms and to prioritize the impli-

cated networks for therapeutic follow-up. Our reanalysis demonstrates that the released evidence does not establish participant-shared alignment of brain activity with context at the psilocybin visit, and that mechanistic or therapeutic inferences based on such alignment may therefore be unwarranted.

Some design safeguards proposed for future studies, including counterbalanced condition order, repeated sampling within each context and evaluation in independent participant cohorts, show potential for testing psychedelic modulation of order-generalizable brain-context relations. Progress in clarifying relations between psychedelic states and context-dependent brain activity requires experimental control and inferential restraint in separating pharmacological, contextual and temporal effects. By neglecting inferential constraints such as separating environmental context from scan position and distinguishing visit associations from therapeutic mechanisms, neuroimaging interpretation drawn from fixed-order psychedelic scans risks a proliferation of unsupported claims about participant-shared brain-context alignment and therapeutic relevance.

Methods

Data and cohorts

De-identified multimodal records from the Article were retrieved from OpenNeuro at accession ds006110 (version 1.2.0)¹¹. Identifiers for 57 participants that were included in the CEBRA re-analysis (that is, contributing all four conditions at both visits at the task-specific modal post-discard run lengths) were determined from the released records. The cohort composition from which 126 participant-visits and 122 directed cross-visit transfers derived was established from released records for 61 participants with both visits and four with baseline data only.

Validation and statistical analysis

The observations were differently partitioned across validation schemes, with within-participant folds ranging from random-frame interleaving to four-fold blocking with a temporal purge, whereas five participant-held-out folds assigned no participant to both training and test data. We tested participant-level CEBRA, TAVRNN, fMRI and EEG contrasts using family-specific procedures implemented in analyses with Benjamini–Hochberg adjustment within each stated family¹². We selected 54 participants who had the lowest psilocybin-visit head motion (average framewise displacement (mm) across four task runs), so as to match numerically the Article’s machine-learning cohort size. Observe that the Article reports a 54-participant machine-learning cohort, but we regard this as specifying a cohort size rather than an MRIQC-selected identity list. We calculated network-replacement diagnostics using fixed-versus-refitted balanced-accuracy contrasts evaluated with 20 size-matched random masks per target under the fixed classifier and participant-level hybrid-to-psilocybin nearest-neighbour distance ratios. To examine validation dependence, we varied the data partition (random-frame, four-fold blocked and five-fold participant-held-out) of the observations and reran classification and latent-geometry analyses. We repeated circular shifts of the intact four-condition sequence 20–100 times per fold across purge gaps of 20, 50, 100 and 150 frames to estimate the location and dispersion of participant-level visit differences in observed-minus-null balanced accuracy under the temporal-null model.

Complementary analyses

The primary TAVRNN outputs that we generated for each of 54 participants at both visits under three model initializations are available in the accompanying code-and-data package (Fig. 2 in the Extended Data). The task-specific absolute off-diagonal effective-connectivity totals used to construct Fig. 3 in the Extended Data were calculated under grey-matter-intersection and sphere-only masks either from participant-paired PEBs or by applying two stacked aggregation specifications using a common posterior-inclusion threshold (> 0.99 ; $n = 62$ rest pairs and $n = 61$ pairs for each other task), and remain available with the accompanying re-analysis data. The complementary analysis is grounded in visit-level fMRI and EEG records (505 volumetric runs, 504 retrievable surface runs and 482 cleaned EEG files) openly released by the Article authors and uses participant-level measures: Fisher- z surface GFC

in 60 complete cases assessed with 10,000 two-sided participant-level sign flips and max- T family-wise correction; strict-motion 332-ROI connectivity and negative-asymmetric modularity¹³ after $FD \leq 0.2$ mm censoring with 264 or more retained frames; paired 64-channel spectra and 8–12-Hz alpha power on linear and decibel scales; and LZ76 counts normalized to temporal sequence length as $C \log_2(N)/N$ ¹⁴.

Data availability

The fMRI, EEG and behavioural data from that Article were obtained from the OpenNeuro dataset ds006110 (version 1.2.0), which was publicly released by the Article authors.

Code availability

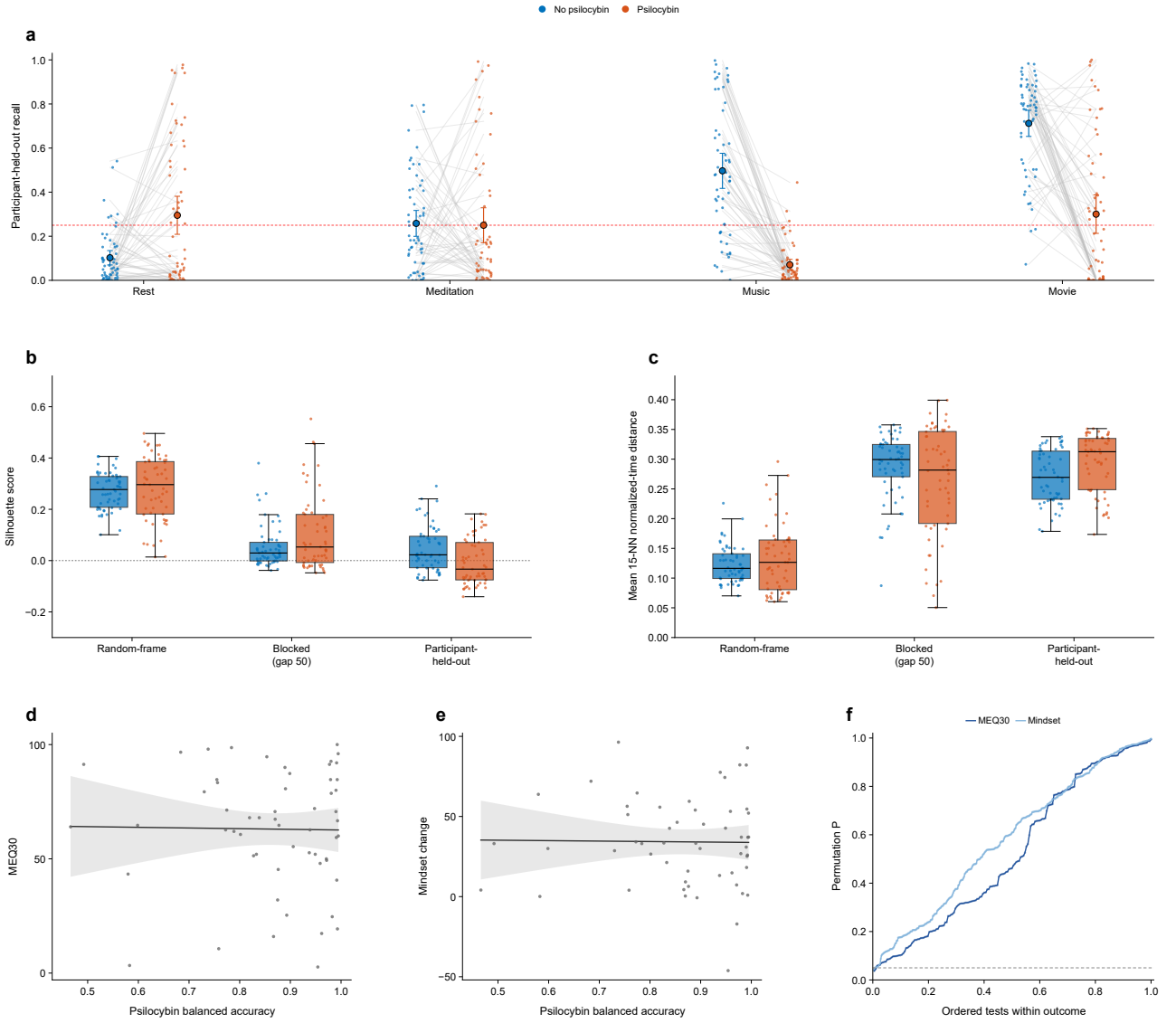
Computational scripts developed to reproduce the results presented here are accessible through <https://github.com/zw-latentmatter/temporal-confounding-reanalysis>.

Competing interests

The author declares no competing interests.

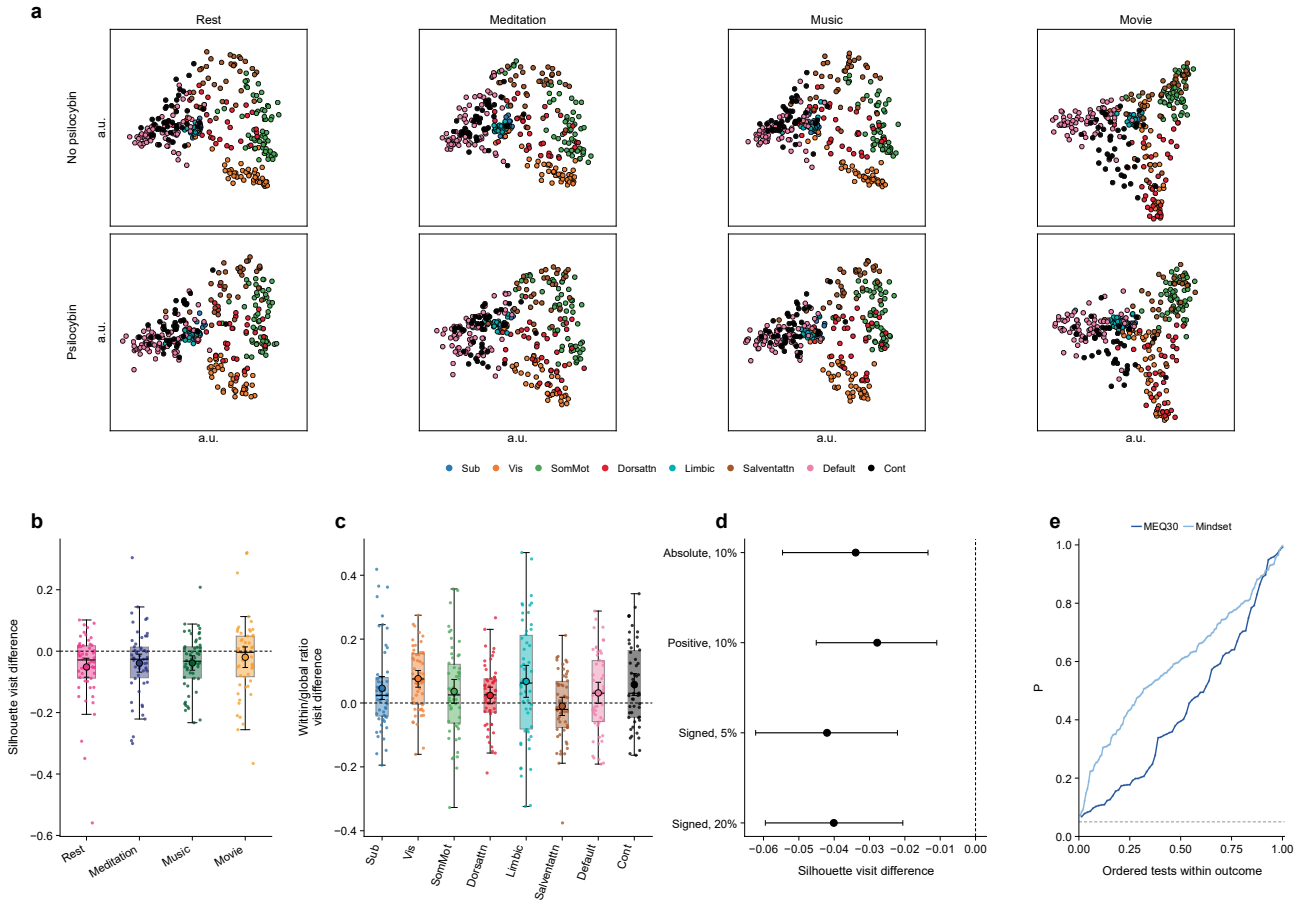
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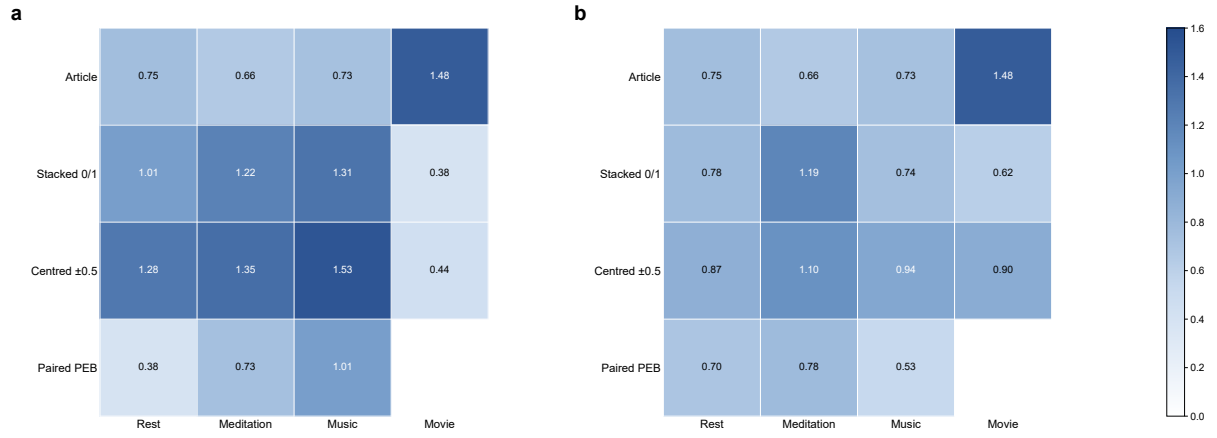
Extended Data Fig. 1 | Participant-level generalization and behavioural assessment of CEBRA estimates under alternative validation. **a**, Participant-held-out recall estimates of $n = 57$ participants with four task classes and two visits, arranged by task. No-psilocybin visits (blue; left within each task) and psilocybin visits (orange; right within each task) are displayed. **b,c**, Participant-level distributions quantifying latent organization among all 57 participants using silhouette scores (condition separation; **b**) and normalized 15-neighbour time distances (temporal locality; **c**). The boxes encode medians and quartile boundaries, whisker limits are the most extreme values within one-and-a-half interquartile ranges, and coloured points denote individual observations ($n = 57$ participants per visit-by-

regime box). **d,e**, Correlation of random-frame psilocybin-visit accuracy per participant with two measures: MEQ30 (**d**) and mindset change (**e**) among available cases. Both were near zero (MEQ30 at $r = -0.015157$; mindset change at $r = -0.013301$), providing no evidence of association ($n = 57$ and 55 , respectively). **f**, Ordered permutation- P values from the complete behavioural analysis (MEQ30, 192 tests; mindset change, 384 tests): for each outcome, permutation- P values were ordered within family; no multiplicity-adjusted association was retained after global correction (minimum $q = 0.995919$; four of 576 values below 0.05; grey dashed line, 0.05 cutoff).



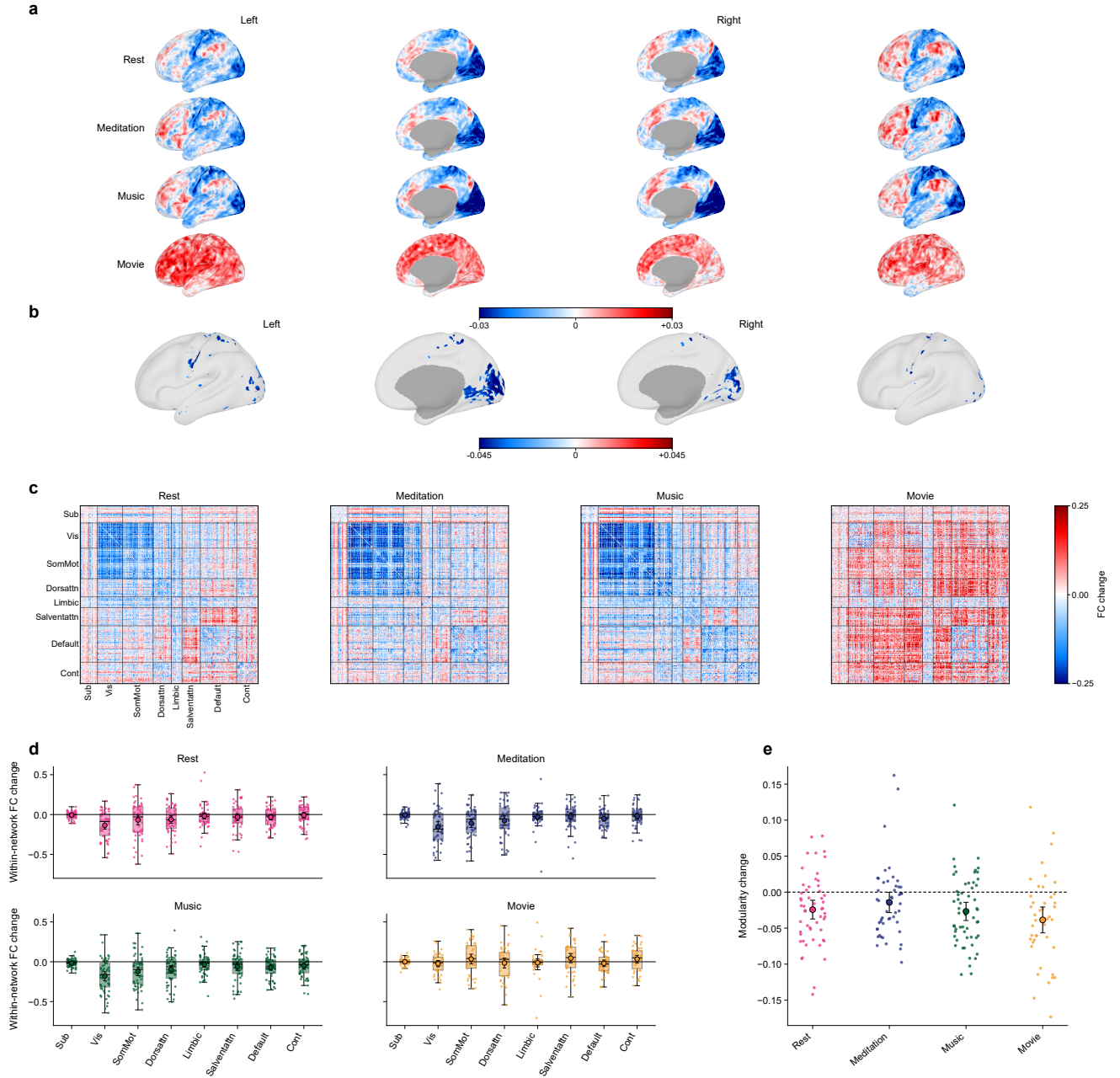
Extended Data Fig. 2 | TAVRNN geometry and robustness following participant-level reanalysis. **a**, Multidimensional-scaling projections of eight visit-task group-mean distance matrices into two dimensions (162 fitted embeddings per projection; Methods). The two rows represent visits that included psilocybin (lower) and corresponding visits which did not include psilocybin (upper); the four columns indicate rest, meditation, music and movie, respectively, with node colours denoting the eight networks. **b**, Distributions of 216 task-specific psilocybin-relative-to-baseline network-silhouette differences from 54 paired participants per task displayed as points, boxes, means and 95% confidence intervals after within-participant visit pairing. **c**, Distributions of 432 participant-level psilocybin-relative-to-baseline

differences in within-network-to-global-distance ratio across eight networks and 54 participants per network. **d**, Forest plot of feature-density controls: mean silhouette differences across the cohort after feature restriction (black points) and at the zero-change reference (dashed line), with 95% confidence intervals shown. **e**, Ordered values of unadjusted P for TAVRNN-behaviour associations, composed of 72 MEQ30 tests and 144 mindset-change tests (grey dashed line, 0.05 cutoff). The coloured curves trace unadjusted P values from two behavioural families, presented to indicate no result below 0.05 (minimum $P = 0.068295$ for MEQ30; minimum $P = 0.071659$ for mindset change; minimum global $q = 0.960285$).



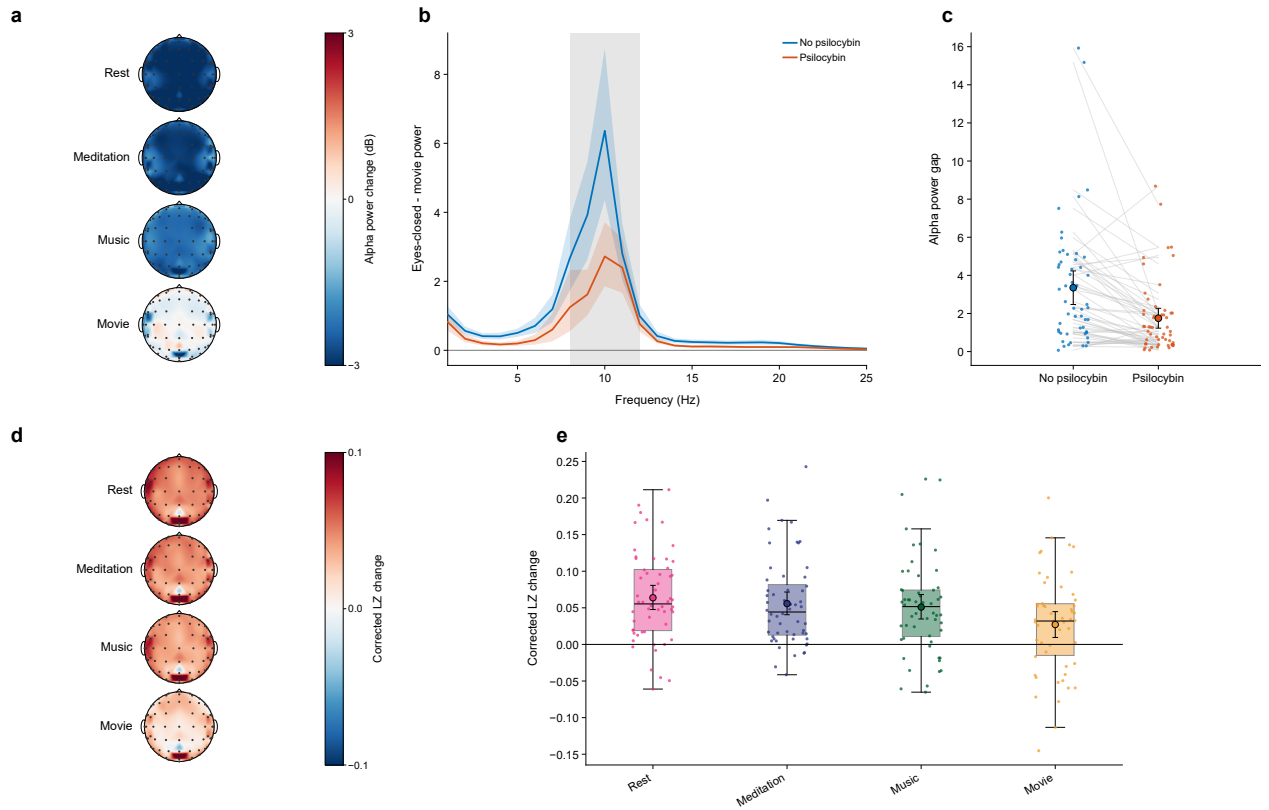
Extended Data Fig. 3 | Comparison of task-resolved DCM totals from the Article (reported reconstruction) to alternative reconstructions. a, Matrix mapping of grey-matter-intersection reconstructions with unsigned effective-connectivity totals. Panel **a** denotes grey-matter-intersection totals

and panel **b** denotes sphere-only totals (in hertz), based on archived summaries aggregated over the cohort using separate mask reconstructions. For the movie condition, the paired-PEB cells represent 0.000 and the Article-reported cells represent 1.48.



Extended Data Fig. 4 | Surface and volumetric fMRI summaries show condition-dependent visit differences under participant-level reanalysis. **a**, Vertex-wise difference in GFC (psilocybin relative to no psilocybin; Fisher- z scale) for every task condition. **b**, Participant-average eyes-closed-minus-movie GFC interaction for every hemisphere in lateral views (outer columns) and across medial views (inner columns). For the 60-participant surface analyses, the eyes-closed-versus-movie interaction (rest, meditation practice and the music task) yielded vertex counts of 30,466, 2,633 and 58,645 after BH FDR, max- T FWER and no thresholding, respectively (coloured vertices remain after max- T FWER control at a five-per-cent family-wise threshold with 10,000 sign flips; grey vertices consequently remained outside the threshold), whereas the unthresholded vertex mean was modestly negative. **c**, Strict-motion participant-paired volumetric 332-ROI FC-change matrices displayed separately for every task, to contrast psilocybin with no-psilocybin visits. **d**, Within-network FC change (psilocybin relative to no psilocybin) exhibited mainly negative estimates across closed-

eye tasks and reduced magnitudes during movie viewing with eyes open. **d,e**, Two marker types for participant-level visit differences (small task-coloured dots) or group means (filled task-coloured symbols with 95% confidence-interval bars) at participant- and group-summary levels after strict-motion censoring. **e**, Modularity comparison across task conditions within paired visits, measured using each participant's paired modularity difference (psilocybin relative to no psilocybin). Average modularity declined significantly between no-psilocybin and psilocybin visits at rest (paired t -test, two-tailed, $q = 6.04 \times 10^{-4}$) and fell between those visits for meditation, music and movie ($1.75 \times 10^{-4} \leq q \leq 0.0473$); the four changes, though necessarily cohort-specific, were consistent in direction. The within-network boxes mark median values and quartile bounds, and whisker reach is one-and-a-half interquartile ranges (paired counts for panels **c–e**: resting state $n = 54$, meditation practice $n = 50$, music listening $n = 56$, movie viewing $n = 44$. One paired visit difference for every participant).



Extended Data Fig. 5 | EEG visit-change summaries under alpha-power and sequence-length-corrected LZ76 estimation. **a**, Alpha-band power spectral density (PSD) contrast between drug-exposed and drug-free sessions. **b**, The corresponding visit-specific eyes-closed-minus-movie mean spectra plotted with 95% bootstrap bands. **c**, Participant-paired alpha-gap distribution after 8–12-Hz power-spectral-density averaging (PSD). **d**, Corrected LZ76 contrast after both sequence-length adjustment and visit subtraction. **e**, The corrected LZ76 contrast between drug-exposed and drug-free visits remained positive across every task. The largest alpha reduction was observed during rest (-2.977 dB, with $n = 56$), but the mean reduction was at least 0.368 dB and statistically detectable in every condition (within-map BH adjustment, $q < 0.05$, minimum 4 of 64 sensors). The analysis

components and corresponding paired cohort counts (participant-level visit differences per component) are: resting-state alpha (panel **a**, $n = 56$), meditation alpha (panel **a**, $n = 56$), music alpha (panel **a**, $n = 56$), movie alpha (panel **a**, $n = 54$), paired spectral-gap analyses (panels **b,c**, $n = 54$), closed-eye corrected LZ76 ($n = 56$ per task) and movie corrected LZ76 ($n = 54$). The corrected-LZ76 boxes mark median values and quartile bounds, and whisker reach is one-and-a-half interquartile ranges. Decibels (dB), hertz (Hz), Benjamini–Hochberg procedure (BH), confidence interval (CI), participant points and within-person lines, Lempel–Ziv 1976 complexity (LZ76), large mean symbols with confidence bars and a grey 8–12-Hz alpha window accompanying power spectral density (PSD).